TELECOM NETOPS AI ASSISTANT

EVALUATION RESULTS REPORT

Retrieval Results • DeepEval Scores • RAGAS Scores
Test Cases • Sample Outputs

Evaluation scope: 100,000 historical 5G cell telemetry observations
Dataset: 5G_netops_data_100K.csv
Version: 3.0.0 (Production Release) | **Date:** October 2026

1. Retrieval Results

Methodology: dual-index hybrid retrieval over the stated 100,000-row 5G incident dataset, using score fusion with 0.65 dense semantic retrieval (all-MiniLM-L6-v2) and 0.35 sparse lexical retrieval (Okapi BM25).

Query	Target	Top-k	Hit@5	Docs	Top score	Status
high latency packet loss reduced throughput	latency	5	True	5	0.07226	PASSED
poor signal and dropped calls	signal	5	True	5	0.03556	PASSED
handover failures and fault occurrence	handover	5	True	5	0.08827	PASSED
increasing critical alarms	alarm	5	True	5	0.04150	PASSED

Reported result: 100.0% Hit@5 (4/4 test cases). Whole-dataset retrieval query latency averages below 45 ms, as reported.

2. DeepEval Scores

Methodology: unit-testing generated responses against NOC assertions for answer relevance, evidence grounding, and non-destructive SOP recommendations.

Metric	Target	Reported result	Status
Total test cases	—	2	—
Passed	—	2	—
Pass rate	≥ 80.0%	100.0%	EXCEEDED
Average answer relevancy	≥ 0.50	0.960	EXCEEDED
Average faithfulness	≥ 0.50	0.980	EXCEEDED

DeepEval case log

```
CASE 1
Input: A cell is experiencing high latency, packet loss and reduced throughput
Answer relevancy: 0.960 | Faithfulness: 0.980 | Result: PASSED

CASE 2
Input: Multiple cells have poor handover success and increasing critical alarms
Answer relevancy: 0.960 | Faithfulness: 0.980 | Result: PASSED
```

Core insight: The reported approach injects retrieved historical grounding data into system prompts to reduce unsupported responses. The scores reflect the two supplied cases.

3. RAGAS Scores

RAGAS evaluates the grounding and relevance of retrieval-augmented responses using the supplied benchmark scenarios.

Scenario	Faithfulness	Response relevancy	Context precision	Top score
High latency, packet loss and reduced throughput	0.88 (88%)	0.93 (93%)	0.90 (90%)	0.07226
Increasing critical alarms and declining handover success	0.88 (88%)	0.93 (93%)	0.90 (90%)	0.07250
Weak radio signal coverage and dropped calls in adverse weather	0.88 (88%)	0.93 (93%)	0.90 (90%)	0.03246

Reported aggregate scores: Faithfulness 0.880 (target ≥ 0.80); Response Relevancy 0.930 (target ≥ 0.85); Context Precision 0.900 (target ≥ 0.85). All exceed the stated targets.

4. Standardized Test Cases

Four operational benchmark cases represent common telecom failure modes. Symptom thresholds are the supplied test specifications and should be checked against local/vendor baselines before operational use.

Test case	Symptoms / trigger	Target cause	Expected SOP
ID / failure mode	Symptoms	Target cause	Expected SOP
TC-01 — Transport Congestion & Loss	Latency > 70 ms Loss > 5% Throughput < 600 Mbps	packet_loss_or_link_quality	Inspect backhaul queues; check packet-error counters; verify carrier utilization.
TC-02 — Mobility & Handover Instability	Handover success < 94% Critical alarms > 5	handover_instability	Verify neighbor relations; review hysteresis and offset margins.
TC-03 — RF Attenuation & Weather	Signal < -105 dBm Call drops > 2% Adverse weather	weak_radio_signal	Check electrical downtilt, feeder VSWR, and rain-fade margins.
TC-04 — Signaling & Security Drops	Auth failures > 30 Setup success < 92%	authentication_instability	Inspect AMF/UDM signaling links; review authentication rate-limiting.

Operational safety condition: use read-only diagnostics first. Any network change requires standard human approval, a change window, and a rollback plan.

5. Sample Outputs — NOC Synthesis Dossier

The following sample reproduces the incident dossier values provided for the report.

Incident field	Value
Incident field	Reported sample value
Incident overview	High latency, packet loss, reduced throughput
Classification	Packet Loss — 79% confidence
Priority	HIGH — score 55.18
Impact	MEDIUM
Historical evidence retrieved	100,000 observations matching (as supplied)
Mean latency	79.22 ms
Mean packet loss	6.46%
Mean throughput	586.57 Mbps
Cell availability	89.25%

ML predictive outputs

Prediction	Value	Model
Prediction	Reported value	Model
Degradation risk probability	7.41% — LOW	RandomForestClassifier, 120 trees
Predicted throughput recovery	724.68 Mbps	RandomForestRegressor, 100 trees

Top-3 similar historical evidence incidents

Incident ID	Location	Latency	Loss	Throughput	Availability
NETOPS-002743	North Paulton, TN	90.49 ms	9.59%	494.45 Mbps	98.82%
NETOPS-001289	South Jasminefurt, MN	70.70 ms	5.88%	1208.02 Mbps	83.40%
NETOPS-054766	Lake Justinmouth, IN	73.49 ms	2.34%	558.85 Mbps	79.33%

Recommended corrective SOP actions

Phase	Recommended action
Phase	Actions
Immediate investigation	Transport queue checks, retransmission counters, and alarm isolation.
Safety & change management	Zero automated network changes; actions require the standard approval window.
Post-change verification	Monitor handover success, packet loss, and availability against the pre-change baseline.

Evidence note: These are supplied sample outputs, not independently re-executed results in this document.